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Handswitch and connector for powered surgical handpiece

Abstract

A handswitch (50) for a powered surgical handpiece (54) comprises a mounting base (62), a lever (70), a spring, a run-safe switch (68) and a lever extension. The mounting base defines a pivot axis. The elongated lever defines a lever axis substantially normal to a direction of the pivot axis. The lever includes a proximal end pivotably connected to the mounting base at the pivot axis and a distal end opposite the proximal end. The first track is on an inner side of the lever. The second track is on an outer side of the lever opposite the inner side, with both tracks substantially parallel to the lever axis. The spring is between the base and the lever and biases the lever about the pivot axis. The run-safe switch is slidably disposed on the first track. The lever extension is slidably disposed on the second track.

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Background/Summary

RELATED APPLICATIONS (1) This application is the National Stage entry of International Patent Application No. PCT/IB2020/053441, filed April 10, 2020, which claims priority to and all the benefits of both U.S. Provisional Patent Application No. 62/832,983, filed on Apr. 12, 2019, and U.S. Provisional Patent Application No. 62/884,765, filed on Aug. 9, 2019, the entire contents of each of which are hereby incorporated by reference.

BACKGROUND

(1) A handswitch for powered surgical handpieces that is easier to install and provides improved convenience of operation is desired.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a perspective view of a handswitch disposed on a handpiece, with the handpiece and handswitch engaged by a connector.

(2) FIG. 2 is an exploded view of the handswitch, handpiece and connector of FIG. 1.

(3) FIG. 3A is a perspective view of a lever of the handswitch of FIG. 2 with a lever extension fully extended.

(4) FIG. 3B is a second perspective view of the handswitch of FIG. 2 with the lever extension fully retracted.

- (5) FIG. 4 is an exploded view of the handswitch of FIG. 2.
- (6) FIG. 5A is a first perspective view of a collar of the handswitch of FIG. 4.
- (7) FIG. 5B is a second perspective view of the collar FIG. 5A.
- (8) FIG. 5C is a top view of the collar of FIG. 5A.
- (9) FIG. 5D is a right-side view of the collar of FIG. 5A.
- (10) FIG. 5E is a front view of the collar of FIG. 5A.
- (11) FIG. 5F is a bottom top view of the collar of FIG. 5A.
- (12) FIG. 5G is a sectional view of the collar of FIG. 5A taken in the direction of arrows 5G of FIG. 5E.
- (13) FIG. 6A is a perspective view of a handpiece housing of FIG. 2.
- (14) FIG. 6B is a right-side view of a part of the handpiece housing of FIG. 6A.
- (15) FIG. 6C is a left-side view of a part of the handpiece housing of FIG. 6A.
- (16) FIG. 6D is an enlarged sectional view of a portion of the housing in circle 6D of FIG. 6B.
- (17) FIG. 7 is a sectional view of the lever and housing of FIG. 1, taken through plane 7 in the direction of arrow 7A.
- (18) FIG. 8A is a perspective view of a lever of the handswitch of FIG. 4.
- (19) FIG. 8B is a top view of the lever of FIG. 8A.
- (20) FIG. 8C is a bottom view of the lever of FIG. 8A.
- (21) FIG. 8D is a right-side view of the lever of FIG. 8A.
- (22) FIG. 8E is a sectional view of the lever of FIG. 8A taken in the direction of arrows 8E of FIG. 8B.
- (23) FIG. 8F is a front view of the lever of FIG. 8A.
- (24) FIG. 9A is a perspective view of a lever extension of the handswitch of FIG. 4.
- (25) FIG. 9B is a top view of the lever extension of FIG. 9A.
- (26) FIG. 9C is a right-side view of the lever extension of FIG. 9A.
- (27) FIG. 9D is a bottom view of the lever extension of FIG. 9A.
- (28) FIG. 9E is a sectional view of the lever extension of FIG. 9A taken in the direction of arrows 9E of FIG. 9D.
- (29) FIG. 9F is a rear view of the lever extension of FIG. 9A.
- (30) FIG. 10A is a perspective view of a switch of the handswitch of FIG. 4.
- (31) FIG. 10B is a top view of the switch of FIG. 10A.
- (32) FIG. 10C is a sectional view of the switch of FIG. 10A taken in the direction of the arrows 10C of FIG. 10B.
- (33) FIG. 10D is a sectional view of the switch of FIG. 10A taken in the direction of the arrows 10D of FIG. 10B.
- (34) FIG. 11 is an exploded view of a first alternative interface between the handpiece and the handswitch.
- (35) FIG. 12A is a perspective view of a handpiece housing illustrating a housing portion of a second alternative interface between the handpiece and the handswitch.
- (36) FIG. 12B is a perspective view of the handpiece housing of FIG. 12A with a handswitch installed illustrating the second alternative interface between the handpiece and the handswitch.
- (37) FIG. 13A is a perspective view of a handpiece and handpiece housing illustrating a housing portion of a third alternative interface between the handpiece and the handswitch.
- (38) FIG. 13B is a perspective view of the handpiece and handpiece housing of FIG. 13A with a handswitch installed illustrating the third alternative interface between the handpiece and the handswitch.
- (39) FIG. 14A is a perspective view of an alternative handswitch.
- (40) FIG. 14B is an exploded perspective view of the handswitch of FIG. 14A.
- (41) FIG. 14C is a sectional side view of the handswitch of FIG. 14A.
- (42) FIG. 15A is a top right front perspective view of a lever of the handswitch of FIG. 14A.

- (43) FIG. 15B is a bottom left front perspective view of the lever of the handswitch of FIG. 14A.
- (44) FIG. 16 is a bottom left front perspective view of a lever extension of the handswitch of FIG. 14A.
- (45) FIG. 17 is a bottom left rear perspective view of a switch of the handswitch of FIG. 14A.
- (46) FIG. 18A is a perspective view of a handpiece and a connector in a connected position.
- (47) FIG. 18B is a perspective view of the handpiece and the connector of FIG. 18A in a released position.
- (48) FIG. 19 is an exploded view of the connector of FIG. 18B.
- (49) FIG. 20A is a sectional side view of the handpiece and the connector of FIG. 18A in a first position taken through plane 20 in the direction of arrows 20A with plane 20 passing through a center axis of the connector.
- (50) FIG. 20B is a broken-out view of a part of the connector shown in a box 20B in FIG. 20A with the connector in a second position.
- (51) FIG. 20C is a sectional side view of the connector of 18B in the first position taken through plane 20' in the direction of arrows 20C with plane 20' passing through the center axis of the connector.
- (52) FIG. 21 is a sectional side view of the connector of FIG. 20 C with a cable installed in the connector.
- (53) FIG. 22 is a perspective view of a cable assembly including the connector of FIG. 18B.
- (54) FIG. 23 is a perspective view of a surgical system including the connector and handpiece of FIG. 18A.

DETAILED DESCRIPTION

- (55) The prior art does not address the need to provide an ergonomically optimized handswitch for a powered surgical handpiece.
- (56) A handswitch for a powered surgical handpiece comprises a mounting base, a lever, a spring, a run-safe switch and a lever extension. The mounting base defines a pivot axis. The elongated lever defines a lever axis substantially normal to a direction of the pivot axis. The lever includes a proximal end pivotably connected to the mounting base at the pivot axis and a distal end opposite the proximal end. The first track is on an inner side of the lever. The second track is on an outer side of the lever opposite the inner side, with both tracks substantially parallel to the lever axis. The spring is between the base and the lever and biases the lever about the pivot axis. The run-safe switch is slidably disposed on the first track. The lever extension is slidably disposed on the second track.
- (57) The handswitch and the features thereof may comprise additional features and modifications as set forth below, such features and modifications being included separately or in combination with each other, with such combinations being limited only by mutual exclusivity.
- (58) At least one of the first track and the second track of a handswitch may be one of a T-slot and a T-track.
- (59) The first track and the second track of a handswitch may both be T-tracks.
- (60) The lever of a handswitch may have first and second lateral sides connecting the inner and outer sides. The run-safe switch of a handswitch may have at least one outboard side extending laterally beyond at least one of the lateral sides of the lever.
- (61) The outboard side of the lever of a handswitch may be defined by an indicator arm.
- (62) The lever of a handswitch may have a slot in the inner side that is in receipt of at least one of a magnet and a magnet retainer fixed to the run-safe switch.
- (63) The slot in the lever of a handswitch may be shorter in length than a length of the lever. The slot may define a travel range of the run-safe switch.
- (64) A detent may be disposed in a handswitch between the run-safe switch and the lever.
- (65) A detent in a handswitch between the run-safe switch and the lever may include a run-safe switch plunger and a first notch associated with an active position of the run-safe switch and a

second notch associated with an inactive position of the run-safe switch.

(66) A handswitch may have the run-safe switch plunger disposed in the lever and the notches disposed on the switch.

(67) A handswitch may have a detent disposed between the lever and the lever extension.

(68) A detent in a handswitch between the lever and the lever extension may include an extension plunger and first, second, third and fourth notches respectively associated with first, second, third and fourth positions of the extension.

(69) The extension plunger of a detent in a handswitch between the lever and the lever extension may be disposed in the lever and the notches may be disposed in the lever extension.

(70) The mounting base of a handswitch may include an annular collar that defines a mounting axis substantially normal to at least the direction of the pivot axis. The collar may be sized for slideable disposition over a proximal end of the handpiece.

(71) The annular collar of a handswitch may define a positioning tab extending axially in a distal direction, the tab including an angled wedge side. The annular collar may further include on an inner diameter thereof one of a detent plunger and a detent notch defining a biasing surface.

(72) The annular collar of a handswitch may include at least a second positioning tab. The second positioning tab may also extend axially in the distal direction and may include an angled wedge side.

(73) The detent plunger of a mounting base detent of a handswitch may be disposed in the collar.

(74) A surgical tool comprises a power surgical handpiece and a handswitch. The power surgical handpiece includes a longitudinally extending handpiece and a magnetic field sensor. The longitudinally extending handpiece housing extends along a housing axis. The magnetic field sensor is disposed in the housing. The handswitch comprises a mounting base, an elongated lever, a spring, a run-safe switch, and a lever extension. The mounting base defines a pivot axis having a direction substantially normal to the housing axis, and a mounting surface extending in a first direction. The elongated lever defines a lever axis substantially normal to the direction of the pivot axis. The lever includes a proximal end, a distal end, a first track, and a second track. The proximal end is pivotably connected to the mounting base at the pivot axis. The distal end is opposite the proximal end. The first track is on an inner side substantially facing the housing axis. The second track is on an outer side opposite the inner side with both tracks substantially parallel to the lever axis. The spring is between the base and the lever and biases the lever about the pivot axis. The run-safe switch is slidably disposed on the first track and has a magnet fixed thereto. The lever extension is slidably disposed on the second track.

(75) The surgical tool and the features thereof may comprise additional features and modifications as set forth below, such features and modifications being included separately or in combination with each other, with such combinations being limited only by mutual exclusivity

(76) At least one of the first track and the second track of a handswitch may be one of a T-slot and a T-track.

(77) The first track and the second track of a handswitch may both be T-tracks.

(78) The lever of a handswitch may have first and second lateral sides connecting the inner and outer sides. The run-safe switch of a handswitch may have at least one outboard side extending laterally beyond at least one of the lateral sides of the lever.

(79) The outboard side of the lever of a handswitch may be defined by an indicator arm.

(80) The lever of a handswitch may have a slot in the inner side that is in receipt of at least one of a magnet and a magnet retainer fixed to the run-safe switch.

(81) The slot in the lever of a handswitch may be shorter in length than a length of the lever. The slot may define a travel range of the run-safe switch.

(82) A detent may be disposed in a handswitch between the run-safe switch and the lever.

(83) A detent in a handswitch between the run-safe switch and the lever may include a run-safe switch plunger and a first notch associated with an active position of the run-safe switch and a

second notch associated with an inactive position of the run-safe switch.

(84) A handswitch may have the run-safe switch plunger disposed in the lever and the notches disposed on the switch.

(85) A handswitch may have a detent disposed between the lever and the lever extension.

(86) A detent in a handswitch between the lever and the lever extension may include an extension plunger and first, second, third and fourth notches respectively associated with first, second, third and fourth positions of the extension.

(87) The extension plunger of a detent in a handswitch between the lever and the lever extension may be disposed in the lever and the notches may be disposed in the lever extension.

(88) The mounting base of a handswitch may include an annular collar that defines a mounting axis substantially normal to at least the direction of the pivot axis. The collar may be sized for slideable disposition over a proximal end of the handpiece.

(89) The annular collar of a handswitch may define a positioning tab extending axially in a distal direction, the tab including an angled wedge side. The annular collar may further include on an inner diameter thereof one of a detent plunger and a detent notch defining a biasing surface.

(90) The annular collar of a handswitch may include at least a second positioning tab. The second positioning tab may also extend axially in the distal direction and may include an angled wedge side.

(91) The detent plunger of a mounting base detent of a handswitch may be disposed in the collar.

(92) A handswitch for a powered surgical handpiece comprises a mounting base, an elongated lever, a spring, a run-safe switch, and a mounting base. The mounting base defines a pivot axis and includes an annular collar defining a mounting axis substantially normal to at least a direction of the pivot axis. The collar is sized for slideable disposition over a proximal end of the handpiece in a direction of the mounting axis. The elongated lever is pivotably connected to the mounting base at a proximal end of the lever. The spring is between the base and the lever, biasing the lever about the pivot axis. The run-safe switch is slidably disposed on the lever. The mounting base further comprises a first positioning tab and a second positioning tab of the collar. Each tab extends axially in a distal direction from a main portion of the collar. Each tab includes an angled wedge side. An inner diameter of the annular collar has disposed thereon one of a detent notch defining a biasing surface and a detent plunger.

(93) The handswitch and the features thereof may comprise additional features and modifications as set forth below, such features and modifications being included separately or in combination with each other, with such combinations being limited only by mutual exclusivity.

(94) Each tab of the handswitch may include an angled wedge side.

(95) Each tab of the handswitch may have a tab centerline about which the tab is substantially symmetrical. Each tab may have two angled wedge sides.

(96) Both tab centerlines of the handswitch may be a predetermined distance below the mounting axis, and on an opposite side of the mounting axis from the pivot axis.

(97) The collar of the handswitch may have the detent plunger disposed in the collar.

(98) The run-safe switch of the handswitch may be slidably disposed on an inner side of the lever.

(99) The handswitch may have a lever extension slidably disposed on an outer side of the lever.

(100) A surgical tool comprises a power surgical handpiece and a handswitch. The power surgical handpiece comprises a longitudinally extending handpiece housing and a magnetic field sensor. The longitudinally extending handpiece housing may extend along a housing axis. The magnetic field sensor may be disposed in the housing. The handswitch comprises a mounting base, an elongated lever, a spring, a run-safe switch. The mounting base defines a pivot axis and includes an annular collar. The annular collar defines a mounting axis substantially normal to at least a direction of the pivot axis. The annular collar is sized for slideable disposition over a proximal end of the handpiece in a direction of the mounting axis. The elongated lever is pivotably connected to the mounting base at a proximal end of the lever. The spring is between the base. The spring biases

the lever about the pivot axis. The run-safe switch is slidably disposed on the lever. The mounting base further comprises a first positioning tab and a second positioning tab of the collar. Each of the tabs extend axially in a distal direction from a main portion of the collar. Each tab includes an angled wedge side. The mounting base further comprises one of a detent notch defining a biasing surface, and a detent plunger, such being disposed on an inner diameter of the annular collar.

(101) The surgical tool and the features thereof may comprise additional features and modifications as set forth below, such features and modifications being included separately or in combination with each other, with such combinations being limited only by mutual exclusivity.

(102) The handswitch may have each tab include an angled wedge side.

(103) Each tab of the handswitch may have a tab centerline about which the tab is substantially symmetrical. Each tab may have two angled wedge sides.

(104) Both tab centerlines of the handswitch may be a predetermined distance below the mounting axis, and on an opposite side of the mounting axis from the pivot axis.

(105) The collar of a handswitch may have the detent plunger disposed in the collar.

(106) The run-safe switch of a handswitch may be slidably disposed on an inner side of the lever.

(107) A handswitch may have a lever extension slidably disposed on an outer side of the lever.

(108) A surgical tool may further comprise two receiving channels and a plurality of contact points. The housing may include the two receiving channels. The channels may be formed on the proximal end of the housing. The receiving channels may be defined at least in part by a peripheral wall and the channels extending axially away from a proximal end of the housing toward a distal end of the housing. A first side of each channel may define a first channel contact point thereof. A second side of each channel defining a second channel contact point thereof. Each positioning tab may extend a second predetermined length in a distal direction. Each tab may include a first side defining a first tab contact point thereof and a second side defining a second tab contact point thereof. When the collar is in engagement with the housing, each first tab contact point is in contact with the first channel contact point of the respective channel, and each second tab contact point is in contact with the second channel contact point of the respective channel. One of a detent plunger and a sloped groove defining a biasing surface may be disposed on an inner diameter of the collar.

(109) The surgical tool and the features thereof may comprise additional features and modifications as set forth below, such features and modifications being included separately or in combination with each other, with such combinations being limited only by mutual exclusivity.

(110) A surgical tool may include a notch in the housing defining the biasing surface and the detent plunger may be disposed on the inner diameter of the collar.

(111) A surgical tool may include the biasing surface facing distally in part. The biasing surface may be engaged by the plunger when the respective contact points of the tabs and the channels are in engagement.

(112) An electrical connector for a surgical tool, the connector comprises an axially extending cable end receiver and an outer sleeve. The axially extending cable end receiver in turn comprises an inner sleeve and a pin holder. The inner sleeve comprises a resilient clip that includes a first retention wedge. The pin holder is fixed to the inner sleeve and comprises a core and a plurality of pins fixed to the core. The outer sleeve is in slidable receipt of substantially an entirety of the cable end receiver. The outer sleeve includes a first window in receipt of the retention wedge. The outer sleeve has a slidable range relative to the receiver. The slidable range extends from a latching position with the wedge in substantial axial alignment with the window, to a releasing position with wedge being axially offset from alignment with the window.

(113) An electrical cable assembly for use with a surgical tool comprises a cable and a connector. The cable comprises a plurality of conductors. The connector comprises an axially extending cable end receiver and an outer sleeve. The axially extending cable end receiver comprising an inner sleeve and a pin holder. The inner sleeve comprises a resilient clip including a first retention wedge. The pin holder comprises a core and a plurality of pins fixed to the core. The pin holder is

fixed to the inner sleeve and is entirely disposed therein. The outer sleeve is in slidable receipt of substantially an entirety of the cable end receiver. The outer sleeve includes a first window in receipt of the retention wedge. The outer sleeve has a slidable range relative to the receiver with the range extending from a latching position with the wedge in substantial axial alignment with the window to a releasing position with wedge being axially offset from alignment with the window. A plurality of the conductors are electrically connected to a plurality of the pins.

(114) A surgical system comprising a handpiece and a cable assembly. The handpiece comprises a housing and an electric drive motor disposed therein. The cable assembly comprises a cable and a connector. The cable comprises a plurality of conductors. The connector comprises an axially extending cable end receiver and an outer sleeve. The axially extending cable end receiver comprising an inner sleeve and a pin holder. The inner sleeve comprises a resilient clip including a first retention wedge. The pin holder comprises a core and a plurality of pins fixed to the core. The pin holder is fixed to the inner sleeve and is entirely disposed therein. The outer sleeve is in slidable receipt of substantially an entirety of the cable end receiver. The outer sleeve includes a first window in receipt of the retention wedge. The outer sleeve has a slidable range relative to the receiver with the range extending from a latching position with the wedge in substantial axial alignment with the window to a releasing position with wedge being axially offset from alignment with the window. A plurality of the conductors are electrically connected to a plurality of the pins.

(115) Each of the above-referenced electrical connector, electrical cable assembly, and surgical system may be refined, i.e., modified, as stated below. Such refinements or modifications may be combined with any of the others without limitation to the extent they are not mutually exclusive.

(116) The cable end receiver may further comprise a bushing disposed between the pin holder and the inner sleeve.

(117) The bushing may include cable engagement feet extending radially inwardly for engagement of the cable.

(118) The bushing may be disposed entirely within the inner sleeve.

(119) The releasing position of the outer sleeve relative to the cable end receiver may be proximal relative to the latching position of the outer sleeve.

(120) In the releasing position, a ramp surface of the wedge may be in engagement with a second surface of the window and a stop surface of the wedge may be spaced from a first surface of the window.

(121) The cable end receiver may further comprise a retaining nut disposed over the inner sleeve and threadably fixed thereto.

(122) The outer sleeve may have a raised alignment surface.

(123) The resilient clip may further comprise a beam fixed to the inner sleeve at both ends, and the wedge may be fixed to the beam between the ends thereof.

(124) The inner sleeve may further comprise a second wedge and the outer sleeve may comprise a second window.

(125) Relative orientations and directions (byway of example, distal, proximal, upper, lower, bottom, rearward, front, rear, back, outboard, inboard, inward, outward, lateral, left, right) are set forth in this description not as limitations, but for the convenience of the reader in picturing at least one embodiment of the structures described. Such exemplary orientations may be from the perspective of a user of the saw.

(126) The elements shown may take many different forms and include multiple and/or alternate components and facilities. The example components illustrated are not intended to be limiting. Indeed, additional or alternative components and/or implementations may be used. Further, the elements shown are not necessarily drawn to scale unless explicitly stated as such.

(127) As illustrated in FIGS. **1-23**, an example surgical tool **50** includes an example handswitch **52**, an example powered surgical handpiece **54**, and an example electrical connector **56**. The example handswitch **52** may be selectively mounted to an example longitudinally extending handpiece

housing **58** of the example powered surgical handpiece **54**. The example connector **56**, selectively connectable to a proximal end **60** of the handpiece **54**, may aid in retaining the handswitch **52** on the housing **58**. When connected to the handpiece **54**, the connector **56** may electrically connect the handpiece **54** via an electrical cable **51** (shown in phantom in FIG. **1**) to a console **53** for communication of control signals and electrical power therebetween.

(128) The cable **51** and the housing **58** may include a signal conductor **55**, e.g., a wire, and a power conductor **57**. The signal conductor **55** may be electrically connected to a magnetic field sensor **59** disposed in the housing **58** and to a controller **63** that may be disposed in the console **53**. The power conductor **57** may be connected to a drive motor **61** disposed in the housing. The drive motor **61** may be powered by any conventional means, e.g., electrical power, pneumatic power. The power conductor **57** may be, by way of example and not limitation, a wire or a pneumatic hose. Each of the conductors **55**, **57** may be comprised of an in-housing portion and an in-cable portion that may connect at a proximal end of the connector **56** or the proximal end **60** of the handpiece **54**. The controller **63** may be programmed to control power to the electric drive motor **61** responsive to signals from sensors, e.g., the magnetic field sensor **59**. The controller **63** may also be electrically connected to other sensors (not shown), with other sensors potentially including switches, e.g., a user-controlled foot switch (not shown). Alternatively, the handpiece **54** may include a battery as a power source, and may communicate signals wirelessly with the console **53**. Yet alternatively, the connector **56** may be a wireless connector and may include one or both of a battery and a wireless communication device, e.g., a dongle.

(129) The handswitch **52** includes an example mounting base **62**, an example lever **64** pivotably connected to the mounting base **62**, an example switch spring **66** disposed between the mounting base **62** and the lever **64** biasing the lever **64** away from the handpiece housing **58**, a run-safe switch **68**, i.e., an unlocked-locked switch **68**, slidably disposed on the lever **64**, and a lever extension **70**, also slidably disposed on the lever **64**.

(130) The base **62** and the handpiece housing **58** cooperatively define an example alignment interface **72**. The run-safe switch **68** and the lever **64** cooperatively define an example run-safe switch interface **74**. The lever extension **70** and the lever **64** cooperatively define an example extension interface **76**. Each of the lever **64**, the lever extension **70**, the switch **68**, the base **62** and the housing **58** include features that define the interfaces **72**, **74**, **76** with such features being described in more detail below. The example interfaces **72**, **74**, **76** may define one or more relative positions between the respective components as is also described in more detail below.

(131) The mounting base **62** may include a first ear **78A** and a second ear **78B**. The ears **78A**, **78B** define a receiving pocket **80**. A proximal tab **82**, which may serve as a visual alignment cue to a user in aligning of the connector **56** with the handpiece **54**, may be provided on a distal end of the mounting base **62** and may define a closure at a proximal end of the pocket **80**.

(132) The ears **78A**, **78B** may be respectively pierced by a first ear aperture **84A** and a second ear aperture **84B**, the apertures **84A**, **84B** aligned with each other on a pivot axis **86** defined by the apertures **84A**, **84B**.

(133) The mounting base **62** may also include as part of the alignment interface **72** an example annular collar **88** that defines a mounting axis **90**. The collar **88** may have an inside diameter **D1** defining an annular mounting surface **91** centered about the axis **90**. The collar **88** may be sized for removable slidable receipt by the proximal end **60** of the housing **58**. An example nominal value of **D1** may be approximately 20 mm (0.80 inches). The pivot axis **86** and the mounting axis **90**, while not intersecting, may be perpendicular to each other when viewed from a direction perpendicular to both axes **86**, **90**, e.g. as shown in FIG. **5C**.

(134) An example first positioning tab **92A**, i.e., a first collar tab **92A**, and an example second positioning tab **92B**, i.e., a second collar tab **92B**, have an axial length **L1** extending from a distal edge **94** of a main, i.e., annular portion **95**, of the collar **88**, to a tip **96A**, **96B** of each of the tabs **92A**, **92B** in a direction parallel to the mounting axis **90**. An example nominal value of **L1** may be

approximately 15 mm (0.60 inches). The tabs **92A**, **92B** may each have a substantially trapezoidal shape when the collar **88** is viewed from the side. The tabs **92A**, **92B**, when viewed from the distal end of the collar **88**, as in FIG. 5E, are arcuately shaped, aligned with and defining in part diameter **D1** of the collar **88**.

(135) The tabs **92A**, **92B** may each be substantially symmetrical about a centerline **98A**, **98B**, respectively, of tabs **92A** and **92B**. For example, the first tab **92A** may have example tapered sides, e.g., a first, i.e., an upper wedge side **100A1** and a second, i.e., a lower wedge side **100A2**, separated by a tab wedge angle of α_1 , and each of wedge sides **100A1** and **100A2** being separated from the centerline **98A** by an angle of $\alpha_1/2$. An example nominal value of α_1 may be approximately 32° , with $\alpha_1/2$ being approximately 16° . Both wedge sides **100A1** and **100A2** are thus angled with respect to centerline **98A**. The distance of the tip from the distal edge **94** to the tip **96A**, i.e., the axial length of the tab **92A**, as noted above, is length **L1**. The tabs **92A**, **92B** may maintain the inside diameter **D1** for their length.

(136) The second tab **92B** may be similarly formed and positioned on the collar **88** and include upper and lower wedge sides **100B1** and **100B2** respectively. Both of the centerlines **98A**, **98B** may be located a distance **B1** below and parallel to the mounting axis **90**, as best shown in FIG. 5D and FIG. 5G. An example nominal value of **B1** may be approximately 2.5 mm (0.10 inches). The wedge sides **100A1**, **100A2**, **100B1**, **100B2** are angled in that each is disposed at an angle to the mounting axis **90**.

(137) The tabs **92A**, **92B** may also define in part a nominal tab contact point **101A1**, **101A2**, **101B1**, **101B2**, one on each of wedge sides **100A1**, **100A2**, **100B1**, **100B2**. The role of the nominal tab contact points **101A1**, **101A2**, **101B1**, **101B2** will be explained in more detail below.

(138) The mounting base **62** may further include a plunger receiving aperture **102** for receipt of an example spring-loaded ball base detent plunger **104**, i.e., a collar detent plunger **104**, best shown in FIG. 7, that comprises part of the alignment interface **72**. The receiving aperture **102** may include a counterbore **106**.

(139) The detent plunger **104** may be commercially purchased. The following description of the detent plunger **104** is exemplary and may be applicable to later-referenced ball plungers. The example plunger **104** may include a substantially constant diameter sleeve **108** formed of steel with a closed end **110** and an open end **112**. The closed end **110** may include a vent aperture. A durable plunger ball **114** (e.g., steel, ceramic) may be disposed in the sleeve **108** to extend, i.e., project, from the open end **112**, with a compression coil plunger spring **116** disposed in the sleeve **108** between the ball **114** and the closed end **110**. The open end **112** may be formed to both provide an inner lip **118** that traps the ball **114** and the spring **116** therein, and provide an outer lip **120** thereat. The outer lip **120** may be used to limit travel of the sleeve **108** into the receiving aperture **102**. The counterbore **106** may be of a depth and diameter to accommodate the outer lip **120**, allowing the plunger **104** to be substantially flush with the mounting surface **91** of the collar **88**, but with the ball **114** being able to extend beyond the mounting surface **91** and past the inside diameter **D1** of the collar **88**. While the plunger **104** has been described as including a sleeve, a coil spring and a ball, the structure described is merely exemplary and alternative plungers may be employed. For example, the plunger may employ a bullet-shaped plunger projectile in place of the ball-shaped plunger projectile. Or a double-ended cylinder, being radiused at one or both ends, may be employed in place of the ball. Alternatively, the plunger may use a spring type other than a coil spring, e.g., a leaf spring, to displace the projectile. Yet alternatively, the plunger may principally comprise an arcuate leaf spring serving as both a spring and a projectile that is received by a recess when aligned therewith. The arcuate leaf spring may be bowed or cantilevered. The plunger may yet alternatively comprise a dome spring in lieu of a separate spring and projectile. The dome spring may be in the form of a resilient thin walled partial hemisphere that may resiliently and substantially elastically deflect at an apex in a compressed condition, inverting an apex portion of the dome, and self-restoring to its partial hemispherical shape, i.e., dome shape, when disposed

over a detent notch. A plunger yet alternatively may rely on magnetic or electromagnet properties of components to bias a projectile in a desired direction and into engagement with a detent notch. (140) An access hole **122** may be formed in a bottom **124** of the collar **88**, opposite of and in alignment with the receiving aperture **102** and the counterbore **106**, to facilitate forming the receiving aperture **102** and the counterbore **106**, and the receipt and installation of the detent plunger **104** in the aperture **102**. The plunger **104** is retained in the receiving aperture **102** by an interference fit, i.e., a press fit, relationship therebetween. A portion of the receiving aperture **102** may be larger than an outside diameter of the plunger **104** to facilitate an aligned receipt of the plunger **104** by the receiving aperture **102**. The access hole **122** may be provided with a chamfer (not shown) to facilitate passage of the plunger **104** therethrough. The plunger **104** is shown in the aperture **102** in FIG. 7.

(141) The alignment interface **72** may include the collar **88**, including the annular portion **95** and the tabs **92A**, **92B** and the plunger **104** and complementary features on the handpiece housing **58**, best shown in FIGS. 6A-6D. Such complementary features may include a first tab receiving channel **128A** and a second tab receiving channel **128B** for receipt, respectively, of the first tab **92A** and the second tab **92B**, and a biasing surface **130** provide by a detent notch **132** defining the biasing surface **130** for engagement by the plunger **104**. The tab receiving channels **128A**, **128B** are positioned on the housing **58** for alignment with and receipt of the collar tabs **92A**, **92B**.

(142) The handpiece housing **58** includes both a grip surface **134** and a relatively recessed engaging surface **136**. The recessed engaging surface **136** is disposed proximal to the grip surface **134**. A peripheral wall **138** defines a border between and connects the two surfaces **134** and **136**. The grip surface **134** may have a first surface diameter **D2**. An example nominal value of **D2** may be approximately 22 mm (0.90 inches). The engaging surface **136** may have a smaller second surface diameter **D3**. An example nominal value of **D3** may be approximately 20 mm (0.80 inches). Both diameters **D2** and **D3** may be centered on a housing axis **140** along which the housing **58** extends. An example height **H1** of peripheral wall **138**, given the example values of **D2** and **D3**, may be approximately 1 mm (0.04 inches).

(143) Diameter **D3** of the engaging surface **136** is smaller than the inside diameter **D1** of the collar **88**, the surface **136** and the collar **88** thereby sized to allow a slideable disposition of the collar **88** over the proximal end of the housing **58** of the handpiece **54** when the housing axis **140** is substantially aligned with the mounting axis **90**. When the handswitch **52** is coupled to the housing **58**, the housing axis **140** may be coincident with the mounting axis **90**. As illustrated in FIG. 5C, the mounting axis **90** is normal to a direction of the pivot axis **86**. The housing axis **140** may accordingly also be normal to the direction of the pivot axis **86** when the handswitch **52** is coupled to the housing **58**. The peripheral wall **138** may extend radially outwardly relative to the mounting axis **90**, connecting the engaging surface **136** with the grip surface **134**. The peripheral wall **138** may define the first tab receiving channel **128A**, i.e., the first channel **128A** and the second tab receiving channel **128B**, i.e., the second channel **128B**.

(144) The first channel **128A** may have an axial length **L2**. More specifically, a channel terminus **142A** analogous to the tip **96A** is spaced the length **L2** from a proximal edge **144** of the peripheral wall **138**. The second channel **128B** may likewise have a length **L2**, and a channel terminus **142B** analogous to the tip **96B** is spaced the length **L2** from the proximal edge **144** of the peripheral wall **138**. An example nominal value of **L2** may be approximately 15 mm (0.60 inches). The axial length **L2** of the channels **128A**, **128B** may be substantially equal to the axial length **L1** of the tabs **92A**, **92B**. The first channel **128A** has upper and lower wedge sides **146A1**, **146A2**. The second channel **128B** has corresponding upper and lower wedge sides **146B1**, **146B2**. The upper and lower wedge sides of both channels have an included angle α_2 therebetween. The angle α_2 may be substantially equal to the angle α_1 . An example nominal value of α_2 may be approximately 34°.

(145) The channel **128A** may be substantially symmetrical about a channel centerline **147A** of the tab receiving channel **128A**. For example, the first channel **128A** may have each of the upper

wedge side **146A1** and the lower wedge side **146A2** separated from the centerline **147A** by an angle of $\alpha/2$. Likewise, the second channel **128B** may have its upper wedge side **146B1** and its lower wedge side **146B2** separated from its channel centerline **147B** by an angle of $\alpha/2$. As noted above, an example nominal value of α may be 34° , with $\alpha/2$ being approximately 17° . The wedge sides **146A1** and **146A2** are thus angled with respect to centerline **147A**, and the wedge sides **146B1** and **146B2** thus angled with respect to centerline **147B**. The channel centerlines **147A** may be located a distance **B2** below the mounting axis **90**, as best shown in FIG. **6B** and FIG. **6C**. The value of **B2** may be substantially equal to the value of **B1**, with an example nominal value of **B2** being approximately 2.5 mm (0.10 inches).

(146) The channel wedge sides **146A1** and **146A2**, together with the tab wedge sides **100A1**, **100A2** define nominal channel contact points **148A1**, **148A2** on the channel wedge sides **146A1**, **146A2** and define the tab contact points **101A1** and **101A2** on the tab wedge sides **100A1**, **100A2**. In an engaged position of the handswitch **52** on the housing **58**, the wedge sides **100A1**, **100A2** respectively engage the channel wedge sides **146A1**, **146A2** at the channel contact points **148A1**, **148A2** that are in contact with the tab contact points **101A1**, **101A2**. Channel contact points **148B1**, **148B2** on the wedge sides **146B1**, **146B2** of the second channel **128B** are similarly defined by engagement of the wedge sides **146B1**, **146B2** with the tab contact points **101B1**, **101B2** of the second tab **92B**. Such engagement occurs either before the tips **96A**, **96B** engage the terminus **142A** of the first channel **128A** or the terminus **142B** of the second channel **128B**, and before the proximal edge **144** engages the distal edge **94** of the collar **88**.

(147) The housing **58** may have a diamond-like carbon (“DLC”) coating. DLC coating may beneficially provide coated surfaces with wear resistance and a lower coefficient of friction.

(148) The collar detent plunger **104** and the detent notch **132** cooperatively define a mounting base detent **149**. The detent **149** may serve to retain the handswitch **52** on the housing **58**, and to bias the tab wedge sides **100A1**, **100A2**, **100B1**, **100B2** against the first channel wedge sides **146A1**, **146A2** and the second channel wedge sides **146B1**, **146B2**. The detent notch **132** may be formed as a recess in the engaging surface **136** and may be symmetrical about a notch centerline **151**. The notch **132** is not required to be symmetrical but may be so if it is formed with a rotating tool, e.g., a mill. The biasing surface **130** of notch **132** is positioned at an engagement angle **3** relative to the mounting axis **90**. An example nominal value of **3** with respect to the housing axis **140** may be approximately 19° . With the mounting base **62** installed on the housing **58**, the plunger's ball **114** is received in part by the notch **132** as shown in FIG. **7**. Deflection of the plunger spring **116** within the sleeve **108** and against the ball **114** presses the ball **114** against the biasing surface **130**, in turn pressing the tab wedge sides **100A1**, **100A2**, **100B1**, **100B2** into engagement with the first channel wedge sides **146A1**, **146A2** and the second channel wedge sides **146B1**, **146B2** at the respective contact points.

(149) The biasing surface **130** is at a proximal end of the detent notch **132** that is engaged by the plunger **104**. The notch may also include a retention edge **150** that may be 90 degrees to the housing axis **140**, or may be radiused, or at an angle greater than the angle $\alpha/3$ and less than 90 degrees. The retention edge **150** may beneficially increase an axial force needed to remove the handswitch **52** from the housing **58**. The edge **150** forces the ball **114** to be suddenly displaced against the plunger spring **116**. The increase in displacement results in an increase in force against the ball **114**, which translates to an increase in force in the axial direction, i.e., in a direction parallel to the housing axis **140**, resisting what may be an unintended removal of the handswitch **52** from the housing **58**.

(150) The tabs **92A**, **92B** and the channels **128A**, **128B** facilitate proper alignment of the handswitch **52** with the housing **58**, in turn facilitating installation of the handswitch **52** on the housing **58**. The combination of the detent **149**, the tabs **92A**, **92B** and the channels **128A**, **128B** allows the handswitch **52** to be properly aligned with the housing **58** for installation of the handswitch **52** thereon, and for retention of the handswitch **52** on the housing **58** prior to

connecting the connector **56** to the housing **58**. An example force required to overcome the detent **149** to remove the handswitch **52** from the housing **58** may be in a range of 27 to 36 Newtons (6 to 8 pounds) of force.

(151) The elongated lever **64**, as stated above, is pivotably connected to the mounting base **62**. The lever **64** defines a lever axis **152** substantially normal to at least the direction of the pivot axis **86** as best shown in FIG. **8C**. The lever **64** includes a proximal end **154** pivotably connected to the mounting base **62** at the pivot axis **86** and a distal end **156** opposite the proximal end **154**. A pivot pin **155**, centered on the pivot axis **86**, may be used to pivotably connect the lever **64** to the base **62**. The lever **64** may have an example nominal overall length **L3** from the distal end **156** to the proximal end **154** of approximately 92 mm (3.6 inches). A first track **158**, i.e., a switch track, is on an inner side **160** of the lever. The inner side **160** of the lever **64** may substantially face the housing axis **140** when installed on a handpiece housing **58**. A second track **162**, i.e., an extension track, is on an outer side **164** opposite the inner side **160** of the lever **64**, with both tracks **158**, **162** substantially parallel to the lever axis **152**. The first track **158** is always located to be, in an installed position, closer to the lever axis **152** than the second track **162**.

(152) The lever **64** may have a width **W1** between opposed first and second lateral sides **165A**, **165B**. An example value of the width **W1** may be approximately 10 mm (0.40 inches). The lateral sides **165A**, **165B** are disposed between and connect the inner side **160** and the outer sides **164**. The lever **64** is biased about the pivot axis **86** about the pivot axis **86** relative to the base **62** by the spring **66**. The run-safe switch **68** is slidably disposed on the first track **158**. The lever extension **70** is slidably disposed on the second track **162**. The lever **64**, like the housing **58**, may be provided with a DLC coating.

(153) The lever **64**, presenting its outer side **164** and the second track **162** in FIG. **8B**, has opposed first and second position indicator wings **166A**, **166B**, respectively, near the distal end **156**. The wings **166A**, **166B** are positioned for alignment with an indicator line **168** on the switch **68**, as described in more detail below. The wings **166A**, **166B** may be labeled with icons **167A**, **167B** indicative of describing the associated switch position. For example, a locked icon **167A** in the form of a schematic representation of a locked, i.e., closed, padlock may indicate a safe, i.e., non-running condition, and an icon **167B** in the form of a schematic representation of an unlocked, i.e., open, padlock may indicate a run condition. Other markings may be used in place of the padlock icons **167A** and **167B**. For example, the words “RUN” and “SAFE” may be employed as icons. The unlocked icon **167B** may be placed closer to the pivot axis **86** than the locked icon **167A**.

(154) An extension plunger aperture **170**, for receipt of a lever extension detent plunger **172**, i.e., an extension plunger **172**, may be formed in the lever **64** on the outer side **164**. The plunger **172** may comprise part of a lever extension multi-position detent. The extension plunger **172** may be spring-loaded, and may be formed like the switch plunger **198**.

(155) A pin aperture **176** for an extension travel limit pin **178**, shown in FIGS. **8B** and **8C**, may be disposed proximate to the distal end **156**, maximizing the extension of the extension **70** relative to the lever **64**. The pin aperture **176** may extend through the lever **64**. A proximal end portion **180** may extend a distance **L4** from the lever's proximal end **154** to an end wall **182** of the proximal end portion **180**. An example value of distance, i.e., length, **L4** may be approximately 18 mm (0.70 inches). The proximal end portion **180** may accommodate a hinge pocket **184** therebeneath, best shown in FIG. **8C** and FIG. **8E**, and may further define a flush outer surface **186** of the proximal end portion **180** that aligns with an outer surface, i.e., a contact surface, **188** of the lever extension **70** when the extension **70** is installed on the lever **64**. The end wall **182** may define a distal end of the flush outer surface **186**.

(156) In FIG. **8C**, the lever **64** presents its inner side **160** and the first track **158**. An unlabeled side of the wings **166A**, **166B** is shown. At the proximal end **154**, the hinge pocket **184** can be seen. In a distal end portion **192**, at the distal end **156**, a tapered end surface **194** is provided. The end surface **194**, when viewed from the side, as shown in FIGS. **8D** and **8E**, may run in a downward, proximal

direction at an angle $\alpha 4$. The angle $\alpha 4$ may be approximately 45° . A switch plunger aperture **196**, for receipt of a switch detent plunger **198**, i.e., a run-safe switch plunger **198** or yet alternatively named a switch plunger **198**, may be formed in the lever **64** on the inner side **160**. The switch plunger **198** may comprise part of a multi-position switch detent and may be spring-loaded, and may be formed like the collar detent plunger **104**. The switch plunger **198** may be shorter than the collar detent plunger **104**.

(157) The aperture **176** for the extension travel limit pin **178** is also shown, with the through aperture **176** facilitating installation of the pin **178**. The aperture **176** may alternatively be a blind hole, not extending all of the way through the lever **64** to the inner side **160**.

(158) The inner side **160** of the distal end portion **192** also may include a switch travel slot **202** that receives a magnet retainer, i.e., a magnet housing **204** which in turn receives a switch magnet **206**, both of which are incorporated into the switch **68** and extend upwardly therefrom. An example length $L5$ of the slot **202** may be approximately 15 mm (0.60 inches). An example width $W2$ of the slot **202** may be approximately 5 mm (0.20 inches). An example depth $H2$ of the slot **202** may be approximately 3 mm (0.12 inches). The switch **68**, the switch magnet **206**, and the magnet housing **204** are shown in FIG. 4. A travel range of the switch **68** relative to the lever **64** may be defined by the length $L5$ of the slot **202**.

(159) The side view of FIG. 8D and the sectional side view of FIG. 8E show that the distal end portion **192** of the lever **64** may have a distal end portion thickness $H3$, greater than a center portion thickness $H4$ of a center portion **207** of the lever **64**. An example value of the distal end portion **192** thickness $H3$ may be approximately 5 mm (0.20 inches), and of the center portion thickness $H4$ may be approximately 3 mm (0.12 inches). The distal end portion **192**, as indicated by the thickness $H3$, may extend a length $L6$ in a proximal direction from the distal end **156** of the lever **64**. An example nominal value of $L6$ may be approximately 40 mm (1.6 inches).

(160) Both of the tracks **158** and **162** can both be seen in FIGS. 8D and 8F, and the second track **162** can be seen in FIG. 8B and the first track **158** can be seen in FIG. 8C.

(161) The first track **158**, i.e., the switch track **158**, can be seen in each of FIGS. 8D and 8F. The first track **158** is included in the distal end portion **192** on the inner side **160**. The first track **158** may be in the shape of a T-shaped track, i.e., a T-track. Alternatively, the track **158** may have a different profile, presenting, for example, a dovetail, a T-slot, a rectangular, or a circular track or slot, or any other appropriate shape. A length $L7$ of the first track **158**, an example length being approximately 38 mm (1.5 inches), may be shorter than the length $L6$ of the distal end portion **192** with a track end **208** that may serve as a switch travel stop for the run-safe switch **68** in an unlocked position, i.e., a run or an active position. Alternatively, the switch travel stop for the run-safe switch **68** in the unlocked position may be provided by a proximal end **210** of the slot **202**, with travel of the switch **68** relative to the lever **64** terminating when the magnet housing **204** engages the proximal end **210** of the slot **202**. Similarly, a switch travel stop for a locked position, i.e., a safe or an inactive position, of the switch **68** may be provided by a distal end **212** of the slot **202**, with travel of the switch **68** relative to the lever **64** terminating when the magnet housing **204** engages the distal end **212** of the slot **202**.

(162) The second track **162**, i.e., the extension track **162**, can be seen in each of FIGS. 8B, 8D and 8F. The second track **162** may extend a length $L8$, from the lever's distal end **156** to the end wall **182** of the proximal end portion **180** on the outer side **164**. An example value of length $L8$ may be approximately 75 mm (3.0 inches). The second track **162** may also be in the shape of a T-shaped track, i.e., a T-track. Like the first track **158**, the second track **162** may alternatively have a different profile, presenting, for example, a dovetail, a T-slot, or circular track or slot, or any other appropriate shape. The end wall **182** may serve as a travel stop for the extension **70** in a stored position. Alternatively, the extension travel stop for the stored position may be provided by a distal end **214** of a pin slot **216** in the extension **70**. Accordingly, travel of the extension **70** to the stored position may be terminated by either engagement of a proximal end **218** of the extension **70** with

the end wall **182**, or by engagement of the distal end **214** of the pin slot **216** by the extension travel limit pin **178**. Similarly, an extension travel stop for the extension **70** in a fully extended position may be provided by a proximal end **218** of the slot **216**, with travel of the extension **70** relative to the lever **64** terminating when the extension travel limit pin **178** engages the proximal end **218** of the slot **216**. Alternatively, the pin **178** may be placed in the extension **70** and the slot **216** in the lever **64**.

(163) The hinge pocket **184** includes opposed first and second side members **220A**, **220B** each with an aligned, co-axial first and second aperture **222A**, **222B** respectively defining or at least aligned with the pivot axis **86** when assembled to the base **62**. Alternatively, the proximal end portion **180** may be formed as a solid mass without the pocket **184**, and with a single aperture passing therethrough on the pivot axis **86**. The apertures **222A**, **222B** may have either a clearance fit or a press fit relationship with regard to the pivot pin **155**. The pin **155** may have a first diameter **D4** on a first end **224** and a second diameter **D5** on a second end **226**, with the first diameter **D4** being greater than the second diameter **D5**. Example nominal values of diameters **D4** and **D5** may be approximately 3.6 mm (0.14 inches) and approximately 3.5 mm (0.13 inches) respectively. The aperture **84A** of the ear **78A** of the mounting base **62** may have a diameter **D6**, an example value of **D6** being equal to or smaller than the diameter **D4** of the first end **224** of the pin **155**, the difference, if any, selected to yield a press fit between the first end **224** of the pin **155**, and the first aperture **84A** of the ear **78A**. The aperture **84B** of the ear **78B** of the mounting base **62** may have a diameter **D7**, equal to or smaller than the diameter **D5** of the second end **226** of the pin **155**, the difference, if any, selected to yield a press fit between the second end **226** of the pin **155**, and the second aperture **84B** of the ear **78B**. The apertures **222A**, **222B** of the side members **220A**, **220B** may both have a diameter **D8** larger than the diameter **D5** of the second end **226** of the pin **155**, the difference selected to yield a clearance fit between the second end **226** of the pin **155**, and the apertures **84B**, **222A**, **222B**.

(164) The switch **68** as shown in FIGS. **10A-10D** may include a switch sliding body **228**. The sliding body **228** may be substantially rectangular and have a switch sliding axis **230**. The sliding axis **230** may be coincident with or substantially parallel to the lever axis **152** when the switch **68** is disposed on the lever **64**. The sliding body **228** has a sliding surface **232** that receives or is received by the first track **158**. Depending on the shape of the first track **158**, the sliding surface **232** may be disposed, by way of example, over or alternatively, inside the first track **158**. An example sliding surface **232** is provided by a T-slot **234** that may extend a length **L9** of the sliding body **228** and may be aligned with the axis **230**. An example nominal value of **L9** may be approximately 21 mm (0.80 inches). Two indicator arms **236A**, **236B** defining corresponding outboard sides **238A**, **238B** extend from opposite sides of the sliding body **228** on opposite sides of the T-slot **234**. Each arm **236A**, **236B** may include the indicator line **168** referenced above. Each indicator line **168** may be formed its associated arm **236A**, **236B** as a recess or a projection. An example width **W4** of the switch **68**, across the outboard sides **238A**, **238B** of the arms **236A**, **236B** may be approximately 21 mm (0.80 inches). Each arm **236A**, **236B** may laterally extend approximately 6 mm (0.24 inches) beyond the lateral sides **165A**, **165B** respectively. An example width **W5** of the lever **64** across the wings **166A**, **166B** of approximately 16 mm (0.63 inches) may also be exceeded by the spread of the arms **236A**, **236B** by approximately 2.5 mm (0.10 inches).

(165) In a bottom surface **240** of the T-slot **234**, a first and a second switch detent recess **242A**, **242B**, comprising part of the switch detent, may be positioned for engagement by a ball (not shown) of the switch plunger **198**. The detent recesses **242A**, **242B** may be hemispherical in shape. The recesses **242A**, **242B** may be separated in the axial direction, i.e., the direction parallel to the axis **230**, by a distance **L10** substantially equal to a distance between the locked icon **167A** and the unlocked icon **167B** on the wings **166A**, **166B**. An example nominal value of **L10** may be approximately 10 mm (0.40 inches). A magnet housing aperture **244** may pass through the bottom surface **240** of the T-slot **234**. The first recess **242A** may be engaged by the plunger **198** when the

switch **68** is in the locked position, i.e., when the indicator line **168** is aligned with the locked icon **167A**. The second recess **242B** may be engaged by the plunger **198** when the switch **68** is in the unlocked position, i.e., when the indicator line **168** is aligned with the unlocked icon **167B**.

(166) The aperture **244** may receive the magnet housing **204** in a press fit relationship. The magnet **206** may be retained in the housing **58** for assembly purposes by petroleum jelly, or alternatively retained by an adhesive or by mechanical entrapment, e.g., a closure, a press fit. A lower side of the switch **68** may have an increased thickness in an area surrounding the aperture **244**. The example magnet **206** and magnet housing **204** may be installed in the switch **68** when the switch **68** is slidably disposed on the lever **64** and the magnet housing aperture **244** is aligned with the switch travel slot **202**.

(167) The lever extension **70** as shown in FIGS. **9A-9F** may include an extension sliding body **246**. The sliding body **246** may be substantially rectangular and have an extension sliding axis **248** extending from a distal end **249** to a proximal end **250** of the extension **70**. The extension sliding axis **248** may be coincident with or substantially parallel to the lever axis **152** when the extension **70** is disposed on the lever **64**.

(168) The sliding body **246** has a lever extension sliding surface **252** that receives or is received by the second track **162**. Depending on the shape of the second track **162**, the sliding surface **232** may be disposed, by way of example, over or alternatively, inside the first track **162**. An example sliding surface **252** is provided by a T-slot **254** that may extend a length **L11** of the sliding body **246** and may be aligned with the axis **248**. An example nominal value of **L11** may be approximately 77 mm (3.0 inches).

(169) The outwardly facing contact surface **188** may be substantially continuous from the distal end **249** to the proximal end **250**. An engagement head **258** may be provided at the distal end **249** of the extension **70**. The engagement head **258** extends upwardly from the contact surface **188** and provides an angled engagement surface **260** for engagement by a thumb or a finger of a user of the tool **50**. A plurality of distal engagement surfaces **262** may be provided in the contact surface **188** proximate to the head **258** for engagement by a user's finger or thumb. The distal engagement surfaces **262** may be defined by recesses or alternatively by projections. The lever extension **70** may similarly have a plurality of proximal engagement surfaces **264** in the contact surface **188** proximate to the proximal end **250**.

(170) In an upper surface **266** of the T-slot **254**, a plurality, e.g., four, detent recesses, i.e., notches, first detent recess **268A**, second detent recess **268B**, third detent recess **268C**, and fourth detent recess **268D**, may be formed therein and positioned for engagement by a ball (not shown) of the extension plunger **172**. The detent recesses **268A**, **268B**, **268C**, **268D** may be substantially hemispherical in shape. The recesses **268A**, **268B**, **268C**, **268D** may be laterally elongated to desensitize a lateral alignment of the recesses **268A**, **268B**, **268C**, **268D** with the plunger **172**. The recesses **268A**, **268B**, **268C**, **268D** may be evenly separated in the axial direction, i.e., the direction parallel to the axis **248**. The first detent recess **268A**, being most proximal recess may be in alignment with the plunger **172** when the extension **70** is in a fully-stored position, e.g., the proximal end of the extension **70** is in engagement with the distal end of the end wall **182**. The fourth recess **268D**, being the most distal recess, may be in alignment with the plunger **172** when the extension **70** is in a fully extended position, e.g., the travel limit pin **178** is engaged by the proximal end **218** of the pin slot **216**.

(171) The pin slot **216** may be formed in the upper surface **266** and may extend an example width **W3** axially for at least an anticipated example travel distance **L12** of the extension **70**. An example nominal value of **W3** may be approximately 2 mm (0.08 inches). An example nominal value of **L12** may be approximately 40 mm (1.6 inches). The pin **178** is received in the slot **216** after the lever extension **70** has been slidably received by the lever **64**. The pin slot **216** may be of a depth **H5** less than a thickness **H6** of the extension. An example nominal value of **H5** may be approximately 1 mm (0.04 inches). An example nominal value of **H6** may be approximately 1.2 mm (0.05 inches).

Having the slot **216** not extend through the entire thickness of the extension **70** is not required for the extension **70** to slide in a desired manner, but may aid in reducing intrusion of debris into the T-slot and an associated increase in resistance to sliding.

(172) The extension **70** may be assembled to the lever **64** by sliding the extension **70** onto the lever **64**, and then inserting the pin **178** through the pin aperture **176** in the lever **64** from the inner side **160** of the lever **64** to extend into the slot **216**. If the pin aperture **176**, as noted above, is provided as a blind hole, a clearance hole (not shown) may be provided in the lever in alignment with the slot **216**. After sliding the extension **70** onto the lever **64**, the clearance hole may be aligned with the pin aperture **176**, and the pin **178** passed through the clearance hole and into the pin aperture **176**. The height of the installed pin **178** must be fit within the available height of the slot **216**. If the slot **216** extends through the extension, the pin **178** should be below the contact surface **188** for the comfort of the user of the handswitch **52**.

(173) In operation, the handswitch **52** is installed on the housing **58** by aligning the tabs **92A**, **92B** with the first channel **128A** and the second channel and axially sliding the collar **88** over the proximal end **60** of the housing **58**. The offset **B1** of the tab centerlines **98A**, **98B** relative to the mounting axis **90** and the offset **B2** of the channel centerlines **147A**, **147B** relative to the housing axis **140** may provide a visual cue to a user that allows faster initial alignment, and helps prevent an incorrect orientation and misinstallation, of the handswitch on the housing **58**. The tabs **92A**, **92B** are received by the first channel **128A** and the second channel. The tab wedge sides **100A1**, **100A2**, **100B1**, **100B2** contact the first channel wedge sides **146A1**, **146A2** and the second channel wedge sides. The wedge sides may cooperatively rotatively align the housing **58** and the handswitch **52** with continued axial movement therebetween to the engaged position, with the handswitch **52** being rotatively aligned with the handpiece housing **58**, i.e., in an aligned position, in the engaged position. As the collar **88** approaches a fully-installed position, the ball **114** of the plunger **104** passes over the edge **150** of the notch **132** and drops into the notch **132**. Once in the notch **132**, the ball **114** is pressed by the spring **116** against the biasing surface **130**, generating a biasing force. The biasing force biases the collar **88**, and thus the handswitch **52** in its entirety, toward a seated position on the housing **58**. In the seated position, the first tab's contact points **101A1**, **101A2** are in contact with the first channel's contact points **148A1**, **148A2**, and the second tab's contact points **101B1**, **101B2** are in contact with the second channel's contact points.

(174) With the handswitch **52** installed on the housing **58**, the connector **56** may be engaged with the housing **58**. The connector **56** may aid in retaining the handswitch **52** on the housing **58**.

(175) In use, the handswitch may have its lever **64** selectively pivoted about the pivot axis **86** towards the housing to operate the tool **50**, e.g., a saw, a bur, a drill, etc. The handpiece **54**, as stated earlier, includes the housing **58**. The handpiece **54** may have disposed within the housing **58** the magnetic field sensor **59**. The magnetic field sensor **59** is in alignment with the switch magnet **206** when the run-safe switch **68** is in the unlocked position and the lever **64** is displaced downwardly towards and potentially against the housing **58**. The sensor **59**, as described above, may be connected to the controller **63** by the signal conductor **55** or wirelessly. Likewise, the motor **61**, disposed in the housing **58**, may be connected to the console **53** and a power source (not shown) therein by the power conductor **57**. Communication of power and control signals between the handpiece **54** and the console **53** may be accommodated by the connector **56**.

(176) Actuation of the tool **50** depends on the field sensor **59** detecting the magnet **206**. The controller **63** may effect a variation in power to a drive motor **61** responsive to a strength of magnetic field detected. Thus, within a sensing range of distance, as the magnet **206** gets closer to the sensor **59**, the power and/or speed of the tool **50** increases. With the switch **68** in the locked position, the magnet **206** is kept out of the sensing range of distance of the sensor **59**, and the motor **61** cannot be actuated.

(177) To operate the tool **50** after the handswitch **52** is installed on the housing **58**, the switch **68** is placed in the unlocked position on the lever. The switch **68** is placed in the unlocked position by

sliding the switch **68** in a proximal direction from the locked position. To do so, the user may engage one or both of the arms **236A**, **236B** with one or more fingers to load the switch **68** in a direction parallel to the switch sliding axis **230**, displacing the switch **68** proximally in the axial direction, i.e., a direction of the axis **230**. The switch **68** may thus be moved to the unlocked position without applying a load to the lever against the switch spring **66**. Sufficient axial force must be applied to the arm or arms **236A**, **236B** to overcome a detent load resulting from engagement of the switch plunger **198** with the recess **242A**. The switch **68** may be axially displaced to the unlocked position by overcoming this load. Travel of the switch **68** in the direction of the unlocked position may be limited by one of engagement of a proximal end of the sliding body **228** with the track end **208**, and engagement of the magnet housing **204** with the proximal end **210** of the slot **202**. Retention of the switch **68** in the unlocked position may be aided by engagement of the plunger **198** with the recess **242B**. No force normal to the sliding axis **230** that might tend to pivot the lever about the pivot axis **86** is required to displace the switch **68** from the locked position to the unlocked position, or to move the switch **68** from the unlocked position to the locked position. The lever **64** is pivoted toward the housing, against the bias of the switch spring **66**. As described above, when the magnet **206** in the switch **68** is within the sensing range, the sensor **59** may provide a signal to the controller **63**. The controller **63** in turn may directly or indirectly provide power to a drive motor **61**, also within the housing **58** to initiate motion of a tool bit **280**. Accidental actuation of the tool **50** is avoided by sliding the switch **68** to the locked position. Release of the lever **64** by the user allows the spring **66** to pivot the lever **64** away from the housing **58**, to an Off position.

(178) The switch **68** may be returned to the locked position by the user by pushing the switch **68** axially in the distal direction. The operation is essentially the same as moving the switch **68** to the unlocked position, but in the opposite direction. The detent load of engagement of the plunger **198** with the recess **242B** is overcome by the user. The switch **68** is slid in the distal direction by the user, with travel of the switch **68** relative to the lever **64** being limited by engagement of the magnet housing **204** with the distal end **212** of the slot **202**. Retention of the switch **68** in the locked position may be aided by engagement of the plunger **198** with the recess **242A**.

(179) The handswitch **52** may be used with the extension **70** in a fully withdrawn position as illustrated in FIGS. **1**, **2** and **3B** a fully extended position as illustrated in FIG. **3A**, or any position in between. There are two intermediate positions illustrated for the example handswitch **52**, but there may be more or fewer intermediate positions. The extension **70** may be retained in any of the exemplary four positions associated with the receipt of the ball of the plunger **172** in a corresponding one of the detent recesses **268A**, **268B**, **268C**, **268D** in the extension **70**. The same detent effect could be achieved by installing the plunger **172** in the extension **70** and providing the detent recesses. The recesses **268A**, **268B**, **268C**, **268D** in the lever extension **70** may each be separated by a distance **L13**, with a total separation between recesses **268A** and **268D** of three times the distance **L13**. An example nominal value of **L13** may be approximately 13 mm (0.50 inches). The recesses **268A**, **268B**, **268C**, **268D** may comprise part of the lever extension multi-position detent. The extension **70** provides a user of the tool **50** with a variable-length lever that can be adjusted to suit a hand size of the user.

(180) When the extension **70** is in the fully withdrawn, i.e., stored, position, the contact surface **188** of the lever extension **70** and flush outer surface **186** of the proximal end portion **180** of the lever **64** cooperatively present a substantially continuous smooth surface for engagement by a user's finger. Retention of the extension **70** in the withdrawn position may be aided by engagement of the extension plunger **172** with the recess **268D**. The user may axially displace the extension **70**, i.e., slide the extension along the axis **248**, by engaging pushing against the extension **70** in a direction of extension travel desired relative to the mounting base **62**. The head **258** provides what may be the most prominent engagement feature of the extension **70** for finger engagement. Supplemental engagement features facilitating gripping of the extension **70** by a finger are provided by the distal

engagement surfaces **262** and the proximal engagement surfaces **264**. Such features **258**, **262**, **264** may resist slippage of the finger across the extension **70**. Sliding the extension **70** from the withdrawn position toward the fully extended position requires that sufficient axial force be applied to the extension **70** in the desired direction of travel to overcome the detent force associated with engagement of the extension plunger **172** with the recess **268D**.

(181) An example withdrawn length of the combined lever **64** and extension **70** may be substantially equal to the sum of **L4** and a distance between the proximal end **250** and the distal end **249** of the lever extension, such a total withdrawn length being approximately equal to 102 mm (4 inches). An example fully extended length of the combined lever **64** and extension **70** may be approximately 142 mm (5.6 inches). The extension **70** may be positioned anywhere between the withdrawn position and the fully extended position. A user may find it helpful to have the extension set to a particular intermediate position. Intermediate detent positions, as may be provided by recesses **268B** and **268C**, may prevent the lever extension translating away from desired intermediate positions along the axis **248** with little or no axial force. The extension **70** as illustrated has the two intermediate detent positions associated with recesses **268B** and **268C**, but this number of intermediate positions may be smaller or greater.

(182) Travel of the extension **70** toward the fully extended position, and away from the end wall **182**, is ultimately limited by engagement of the pin **178** with the proximal end **218** of the slot **216**. Retention of the extension **70** in the fully extended position may be aided by engagement of the extension plunger **172** with the recess **268A**. Return of the extension **70** to the withdrawn position may be achieved by pulling the extension **70** axially in the proximal direction until travel is limited by one of engagement between the proximal end **250** of the extension **70** against the end wall **182**, and engagement of the pin **178** against the distal end **214** of the slot **216**. Retention of the extension **70** in the withdrawn position may again be aided by engagement of the extension plunger **172** with the recess **268D**.

(183) In any extended position, i.e., any position other than the withdrawn position, the extension presents a significant contact pad, including the outer surface **188** and the head **258**, separate from the proximal end portion **180**, for engagement by a user's finger. The contact pad may present a width of the extension, e.g., approximately 10 mm (0.40 inches) and an approximate length of 85 mm (3.3 inches). Such a large area contact area, facilitated by locating the extension **70** on the track **162** on the outer side **164** of the lever **64**, improves the comfort of the user's finger when depressing the extension **70** to operate the tool **50**.

(184) FIGS. **11** through **13B** illustrate alternative mounting base configurations.

(185) FIG. **11** shows a handpiece housing **58'** and a mounting base **62'** with just a single positioning tab **92'**. Wedge sides **100'** of the tab **92'** engage wedge sides **146'** of a single receiving channel **128'**. Although not illustrated, the base **62'** and the housing **58'** may include a detent like the mounting base detent **149** for retention of the handswitch **52'** on the handpiece housing **58'**. A contact point may also be provided between a peripheral wall **138'** of the housing **58'** and a distal edge **94'** of a collar **88'** at a bottom of the collar **88'**.

(186) FIGS. **12A** and **12B** show a handpiece housing **58''** with a receiving channel **146''** that slideably receives a handswitch **52''** with a mounting base **62''**. The mounting base **62''** has retention tangs **92''** with barbs **100''**. The tangs **92''** are spring-biased to spread outward. When the barbs **100''** pass an end of the channel **146''**, the tangs **92''** spread, and the barbs **100''** engage a proximal end **60''** of the housing **58''** on either side of the channel **146''** to retain the handswitch **52''**.

(187) FIGS. **13A** and **13B** show a handpiece housing **58'''** with a receiving channel **146'''** having a dovetail profile and disposed transversely to a housing axis **140'''**. A handswitch **52'''** has a mounting base **62'''** with a rail **92'''** having a dovetail shape complementary to that of the channel **146'''**. The channel **146'''** has a detent notch **132'''** therein for engagement by a plunger (not shown) disposed in the mounting base **62'''**. The rail **92'''** of the mounting base **62'''** is received by the

channel **146'''** and is retained therein by engagement of the plunger with the detent notch **132'''**.

(188) A yet alternative handswitch **52'''** having an alternative lever **64'''**, switch spring **66'''**, run-safe switch **68'''** and lever extension **70'''** is shown in FIGS. **14A-17**. The lever **64'''** may be received by a receiving pocket **80'''** of a mounting base **62'''**. A pivot pin **155'''**, centered on a pivot axis **86'''**, may be used to pivotably connect the lever **64'''** to the base **62'''**. The mounting base **62'''** may also include an example annular collar **88'''** that defines a mounting axis **90'''**.

(189) The lever extension **70'''** is slidably disposed on the lever **64'''** for selective translation by a user between a fully withdrawn position, shown in FIG. **14A** and FIG. **14C**, and a fully extended position, not shown, but analogous to the fully extended position of the lever extension in FIG. **3A**. The switch **68'''** is slidably disposed on the lever **64'''** for selective translation by a user between a unlocked position and a locked position.

(190) The lever **64'''** defines a lever axis **152'''** substantially normal to at least a direction of the pivot axis **86'''**. The lever extension **70'''** and the run-safe switch **68'''** may each have a respective sliding surface **252'''**, **232'''** aligned with the lever axis **152'''**.

(191) The lever **64'''** has a first track **158'''** and a second track **162'''** extending substantially parallel to the lever axis **152'''**. The first track **158'''** is located, in an installed position, closer to the mounting axis **90'''** than the second track **162'''**. The second track **162'''** is on an outer side **164'''**. The first track **158'''** may include an interior inward-facing, i.e., an interior inner side **160'''**, of the lever **64'''**. Neither the inner side **160'''** nor the outer side **164'''** are required to be exterior surfaces of the lever **64'''**. The lever **64'''**, being hollow, may have multiple inner sides, e.g., an interior inner side **1632'''**, and an exterior inner side **163'''**, most proximate to the axis **90'''**. The lever **64'''** may likewise have multiple outer sides.

(192) The lever **64'''** may be formed of sheet metal by stamping. The lever **64'''** may include the switch spring **66'''** as an integral part of the lever **64'''**, formed integrally therewith, for disposition between the rest of the lever **64'''** and the base **62'''**. The spring **66'''** may bias the lever **64'''** about the pivot axis **86'''**. The lever **64'''** may include a plurality of (e.g., two) joining tabs **282'''** at a first seam edge **284'''** having a first shape, each of the tabs **282'''** being received by a joining notch **286'''** at a second seam edge **288'''** of the lever **64'''**. An example shape for the tabs **282'''** and the notches **286'''** may be a dovetail as illustrated.

(193) The lever **64'''** may include a downwardly extending switch-stop tab **204'''** and an upwardly extending extension-stop tab **178'''**. The stop tabs **204'''**, **178'''** may be used to limit travel of the switch **68'''** and the lever extension **70'''** respectively as described in more detail below.

(194) The lever **64'''** may also include a switch plunger **198'''** and an extension plunger **172'''**. The extension plunger **172'''** may be formed in the outer side **164'''** of the lever **64'''**, and may be in the form of a cantilevered leaf spring **290'''** with an integral projection **292'''** disposed at an end thereof. The switch plunger **198'''** may be formed in the exterior inner side **163'''** of the lever **64'''**. The switch plunger **198'''** may be in the form of a cantilevered leaf spring **296'''** with a projection **298'''** integral therewith disposed at an end thereof.

(195) The lever extension **70'''** is slidably disposed on the second track **162'''**. The sliding surface **252'''** of the lever extension **70'''** may receive the second track **162'''**.

(196) An extension travel slot **216'''** may be formed in an upper surface **266'''** of the sliding surface **252'''**. The slot **216'''** may extend for at least an anticipated example travel distance of the extension **70'''**. The extension-stop tab **178'''** is received in the slot **216'''** after the lever **64'''** has been slidably received by the lever extension **70'''**. The tab **178'''** may elastically deflect upon initial receipt of the lever **64'''** by the sliding surface **252'''** of the lever extension **70'''**. When the extension **70'''** is pushed over the lever **64'''** a sufficient distance, the tab **178'''** is received by the slot **216'''** and snaps back to extend into the slot **216'''**.

(197) A plurality, e.g., three, detent recesses, i.e., notches, **268'''** may be formed in the upper surface **266'''** and positioned for engagement by the projection **292'''** of the extension plunger **172'''**. The detent recesses **268'''** may be substantially semi-cylindrical in shape, and may extend

laterally across substantially a full width of the upper surface **266'''**. The recesses **268'''** may be evenly separated in the axial direction, i.e., the direction parallel to the axis **152'''**, or, as illustrated in FIG. **14C** and FIG. **16**, may be spaced unevenly. A most distal of the recesses **268'''** may be in alignment with the plunger **172'''**, and more particularly the projection **292'''** thereof, when the extension **70'''** is in a fully-stored position. A most proximal of the recesses **268'''**, may be in alignment with the plunger **172'''**, i.e., the projection **292'''** when the extension **70'''** is in the fully extended position, e.g., the fully extended position being achieved when the tab **178'''** is engaged by a proximal end **218'''** of the slot **216'''**.

(198) Two detent recesses **242'''** may be formed in a bottom surface **240'''** of the sliding surface **232'''** of the switch **68'''**. The recesses **242'''** may be positioned for engagement by the projection **298'''** of the switch plunger **198'''**. The detent recesses **242** may be semi-cylindrical in shape and may extend laterally across an entire width of the switch **68'''**, as illustrated in FIG. **17**. The recesses **242'''** may be separated in the axial direction, i.e., the direction parallel to the axis **152'''**, by a distance substantially equal to a distance between locked and unlocked positions of the switch **68'''**. A more distal of the recesses **242'''** may be engaged by the projection **298'''** when the switch **68** is in the locked position. A more proximal of the recesses **242'''** may be engaged by the projection of the plunger **198'''** when the switch **68'''** is in the unlocked position. A magnet **206'''** may be press fit in a magnet aperture **244'''** in the switch **68'''**.

(199) An upper surface **300'''** of the sliding surface **232'''** of the switch **68'''** may include a switch travel slot **202'''** formed therein. An axial length of the slot **202'''** may be determine a travel range of the switch **68'''** relative to the lever **64'''**. The switch-stop tab **204'''** is received in the slot **202'''** after the switch **68'''** has been slidably received by the first track **158'''** of the lever **64'''**. The tab **204'''** may elastically deflect upon initial receipt of the sliding surface **232'''** of the switch **68'''** by the first track **158'''** of the lever **64'''**. When the switch **68'''** is pushed into the lever **64'''** a sufficient distance, the tab **204'''** is received by the slot **202'''**.

(200) The connector **56** may be beneficially configured to facilitate engagement with and release from the handpiece **54**. FIGS. **18A-23** show an example connector **56**. The connector **56** may be connected to a first end of the cable **51**. The connector **56** and the cable **51** may be included as elements of an electrical cable assembly **301**. The cable assembly **301** may further comprise a second connector **302** fixed to a second end of the cable **51**. The second connector **302** is illustrated schematically, and may or may not be identical to the connector **56**. The cable assembly **301** and the handpiece may collectively comprise a surgical system **303**.

(201) The connector **56** includes an axially extending cable end receiver **304** and an outer sleeve **305**. The cable end receiver **304** may be fixed directly to an end of the cable **51**. The outer sleeve **305** receives and is disposed over substantially an entire length of the cable end receiver **304**.

(202) The cable end receiver **304** includes a pin holder **306** that is disposed within an inner sleeve **308** and is fixed thereto. A bushing **310** may be disposed against a proximally facing side **312** of the pin holder **306** to retain or aid in retaining the pin holder **306** inside of the inner sleeve **308**. A retaining nut **314** may be disposed over a proximal end **316** of the inner sleeve **308** to aid in the retention of the pin holder **306** within the inner sleeve **308**. A washer **318** may be coaxially disposed between the nut **314** and the inner sleeve **308**. The outer sleeve **305** extends over substantially an entirety of the receiver **304**, including the nut **314** if included, such that the receiver **304** cannot be gripped by a user to pull the connector **56** from the handpiece **54**. The connector **56** can only be removed from the handpiece **54** by gripping the outer sleeve **305** and then pulling away from the handpiece **54**, i.e., in a proximal direction. The constituent elements will be described in more detail below.

(203) The outer sleeve **305** is substantially cylindrical and includes a gripping portion **319** and an insertion portion **320**. The insertion portion **320** may be of a substantially constant first outside diameter and a substantially constant first inside diameter, both sized to be received by a handpiece electrical receptacle **321**. The receptacle may define an annular gap between a center boss and a

surrounding inside diameter surface. The insertion portion **320** may also include one or more windows **322** for receipt of a corresponding number of retention wedges **330** disposed on the inner sleeve **308**. The illustrated insertion portion **320** shows two such windows **322**. The windows **322** are shown as being oppositely disposed across the axis **140** from each other. Each of the windows **322** may have a first surface, e.g., a proximal engagement surface **323** and a second surface, e.g., a distal engagement surface **324** for respective engagement with a distally facing ramp surface **325** of the wedge **330** and a proximally facing stop surface **326** of the wedge **330**.

(204) The gripping portion **319** of the outer sleeve **305** may include at a raised alignment surface **328**. The alignment surface **328** may present as a lobe, e.g., a raised surface extending radially further from the axis **140** than an oppositely disposed surface of the outer sleeve **305**, that may align with, for example, the proximal tab **82** of FIG. 5C, to facilitate alignment with the handpiece **54**. Such alignment also yields alignment of pins **332** of the connector **56** with receiving sockets **334** within the receptacle **321** of the handpiece **54**, and alignment of the wedges **330** with engagement recesses **335** disposed in the inside diameter surface of the annular gap of the receptacle **321**. The gripping portion **319** may also include gripping surfaces **336**, e.g., a plurality of parallel recesses and a V-shaped surface pocket as shown in FIG. 22, or a plurality of raised edges on sides thereof. A length **L14** of the outer sleeve **305** is sufficient to substantially cover the assembled inner sleeve **308**, pin holder **306**, bushing **310**, washer **318** and nut **314**.

(205) The inner sleeve **308** includes a shank portion **338** and a base portion **339**. The shank portion **338** is substantially entirely disposed within the insertion portion **320** of the outer sleeve **305**. The wedges **330** may be mounted to the shank portion **338**. More specifically, each wedge **330** may be fixedly mounted to an associated resilient and elastically deflectable beam **340**. Together, the wedge **330** and the beam comprise a resilient clip. Each beam **340** and wedge **330** may be formed integrally with the shank portion as a single unitary forming. The beams **340** may be longitudinally oriented parallel to the axis **140** on opposite sides of the axis. The beams **340** may be defined by longitudinal slits in the shank portion **338**, and may have each of a distal end and a proximal end disposed within and fixed to the shank portion **338**. The beams **340** may have a thinner cross-section thickness than the rest of the shank portion **338** to facilitate deflection. The base portion **339** is substantially entirely disposed within the gripping portion **319** of the outer sleeve **305**. The inner sleeve **308** is able to axially translate a limited distance within the outer sleeve **305**. The inner sleeve **308** includes an inner chamber **341** that may receive each of the pin holder **306** and the bushing **310**. The pin holder **306** and the bushing **310** are both fixedly inserted into the inner chamber **341**.

(206) The pin holder **306** includes a pin holder core **342** and a plurality of the pins **332** mounted within and fixed to the pin holder core **342**. A locator ring **344** may be formed as an integral unitary portion of the core **342**. The pin holder core **342** may be made of a non-conductive material, e.g., rubber, plastic. The pins **332** may be made of a rigid conductive material, e.g., metal.

(207) A distal portion of the pin holder **306** is distal to the locator ring **344** and may be disposed within the shank portion **338** of the inner sleeve **308**. A proximal portion of the pin holder **306** is proximal to the locator ring **344** and may be disposed within the base portion **339**. A plurality of pins **332**, e.g., nine, may be disposed in parallel to the axis **140** in the pin holder **306**, with tips of the pins **332** extending beyond the core **342** on a pin distal end **346** and wire receiving pockets disposed on the proximal end **348** of the pins **332**. The core **342** may be hollow, e.g., have a cavity **350** open on a proximal end thereof.

(208) The bushing **310** may have an interior chamber that extends a length of the bushing **310**. The bushing **310** may have an opening at a distal end **352** sized to receive the proximal end of the pin holder **306**. Travel of the bushing **310** relative to the pin holder **306** in the axial direction within the inner sleeve **308** may be limited by the locator ring **344** of the core **342**. The travel of the bushing **310** may be so limited by engagement between the proximally facing side **312** of the pin holder **306**, e.g., a proximal side of locator ring **344** serving as a pin holder core locator surface and the

distal end **352** of the bushing **310**.

(209) The locator ring **344** may include a keyway notch **354** for receipt of a keytab **356** formed on the bushing **310**. Alignment of the keytab **356** and the keyway notch **354** ensure alignment of the bushing **310** with the pin holder **306**.

(210) The bushing **310** may also include a plurality of cable engagement feet **358**, each disposed at the end of a bushing leg **360**. The bushing legs **360** may be defined by a corresponding plurality of slots **362** in a proximal end **364** of the bushing **310**. The bushing legs **360** may define a rake edge **366** for engagement with a rake notch **368** on an inside diameter of the base portion **339** of the inner sleeve **308**. The engagement between the rake edge **366** and the rake notch **368** aids in retaining each of the bushing **310** and the pin holder **306** in the inner sleeve **308**. The cable engagement feet **358** are located at the proximal end **364** of the bushing **310** and are orientated to face radially inwardly. The feet **358** may engage the cable **51** when the connector **56** is fixed to the cable **51**.

(211) The nut **314** includes a threaded cylindrical body **370** and a proximal end flange **372** at a proximal end of the cylindrical body **370**. The nut **314** includes threads on an inner diameter of the threaded cylindrical body **370** that may be in threaded engagement with threads on an outer diameter of the proximal end **316** of the inner sleeve **308**. A distal end of the nut **314** may engage a proximally-facing engagement surface of the base portion **339** of the inner sleeve **308**, limiting travel of the nut **314** thereon. A plurality, e.g., five, spanner pin apertures **374** may be formed in the end flange **372**. The spanner pin apertures **374** may receive the engagement pins (not shown) of a spanner wrench (not shown).

(212) The washer **318** may be disposed inside the nut **314**, between the proximal end **316** of the inner sleeve **308** and the end flange **372**.

(213) As part of an assembly process, the cable **51** may be connected to the pins **332** within the pin holder **306**. Individual conductors **55** of the cable **51** may each be placed in a pre-determined pin pocket of the distal end **346** and fixed to, e.g., soldered to, the associated pin **332**. After the conductors **55** of the cable **51** are connected to the pins **332**, the pin holder **306** and the bushing **310** may be placed into or connected to a mold (not shown) that encloses an end portion of the cable proximal to the bushing **310**. An insulator **376** having an uncured liquid form, e.g., silicone, may be injected into the mold, and thus into the connector **56**, sealing the ends of the pins **332** disposed inside the bushing **310** and over the cable **51**. The cured insulator **376**, i.e., the overmolding **376**, is solid. The overmolding **376** may be elastically flexible, allowing a portion of the cable over which the overmolding **376** is disposed, to flex. The overmolding **376** may protect the pins **332** on the end of the pin holder **306** in the connector **56** from contamination.

(214) The washer **318** may be disposed inside the nut **314**, between the end flange **372** and the proximal end **316** of the inner sleeve **308**. The nut **314** may be threaded over the inner sleeve **308** after forming the overmolding **376**. Thus, the washer **318** may be pressed against the end flange **372** by the overmolding **376**.

(215) The assembled cable end receiver **304**, i.e., the inner sleeve **308**, pin holder **306**, bushing **310**, nut **314**, washer **318**, and cable **51**, may be axially and rotatively aligned with the outer sleeve **305** and pressed thereinto. The inner sleeve **308** may have a guide rail **378** that is received by a complementary guide slot **380** to ensure axial alignment of the wedges **330** with the windows **322**.

(216) When the connector **56** is fully assembled, the outer sleeve **305** extends over substantially the entire combined inner sleeve **308** and nut **314**.

(217) In operation, the connector **56** in an uninserted condition has the wedges **330** extending radially outwardly through the windows **322** beyond the first outside diameter of the insertion portion **320**. In the uninserted condition, the outer sleeve **305** is in a latching position with respect to the inner sleeve **308**. In the latching position, the wedges **330** are in substantial axial alignment with the windows **322** of the outer sleeve **305**, allowing the beams **340** to be substantially undeflected and the wedges **330** to extend beyond the first outside diameter of the insertion portion

320. The connector **56** is aligned with the handpiece **54** using the alignment surface **328** to orient the connector **56** with respect to the handpiece **54**. The connector **56** is then pushed into engagement with the handpiece **54**. During insertion, engagement of the ramp surface **325** of the wedges **330** against an inside diameter of the receptacle **321** of the handpiece **54** biases the wedges **330** down into the windows **322**, against the resistive force of the beams **340**. The inner sleeve **308** and the outer sleeve **305** remain in the latching position, with the wedges **330** aligned with the windows **322**. As the windows **322** come into axial alignment with the engagement recesses **335** in the handpiece **54**, the wedges **330** are restored to the extended position by the elastic beams **340**. The wedges **330** are resultantly disposed in part in the recesses **335**.

(218) The wedges **330** tend to resist withdrawal of the connector **56** from the handpiece **54**. However, with application of sufficient withdrawal force to the outer sleeve **305**, the resistance to withdrawal may be overcome by the action of the distal engagement surfaces **324** of the windows **322** acting on the ramp surfaces **325**. The force against the ramp surfaces **325** first pushes the stop surfaces **326** of the wedges **330** against a proximal end of the recess **335**, restricting movement of the inner sleeve **308** relative to the handpiece **54**. With an increase in the withdrawal force applied to the outer sleeve **305**, the distal engagement surfaces **324** of the windows **322** acting against the ramp surfaces **325**. The force on the outer sleeve **305** overcomes the resistance of the beams **340** to displace the wedges **330** radially inwardly. Such displacement of the wedges allows the outer sleeve **305** to move axially relative to the inner sleeve **308** to a releasing position therebetween with the wedges **330** being axially offset from alignment with the windows **322**. With the outer sleeve **305** in the releasing position relative to the inner sleeve **308** as shown in FIG. 20B, the wedges **330** are withdrawn, i.e., removed, from the recesses **335**. Such removal of the wedges **330** from the recesses **335** allows the connector **56** to be withdrawn from the receptacle **321**. Once removed from the handpiece **54**, the connector **56** may be restored to the latching position by a restorative force of the beams pressing the ramp surfaces **325** against the distal surfaces **324** of the windows **322** to produce relative axial movement across a slidable range between the inner sleeve **308** and the outer sleeve **305**.

(219) Pulling on the cable **51** alone will not result in a release of the connector, as the stop surfaces **326** will engage the proximal end of the recesses **335**. Accidental removal of the connector **56** is thus resisted.

(220) In the drawings, the same reference numbers indicate the same elements. Further, some or all of these elements could be changed. With regard to the media, processes, systems, methods, heuristics, etc. described herein, it should be understood that, although the steps of such processes, etc. have been described as occurring according to a certain ordered sequence, such processes could be practiced with the described steps performed in an order other than the order described herein. It further should be understood that certain steps could be performed simultaneously, that other steps could be added, or that certain steps described herein could be omitted. In other words, the descriptions of processes herein are provided for the purpose of illustrating certain embodiments, and should in no way be construed so as to limit the claims.

(221) Accordingly, it is to be understood that the above description is intended to be illustrative and not restrictive. Many embodiments and applications other than the examples provided would be apparent upon reading the above description. The scope should be determined, not with reference to the above description, but should instead be determined with reference to the appended claims, along with the full scope of equivalents to which such claims are entitled. It is anticipated and intended that future developments will occur in the technologies discussed herein, and that the disclosed systems and methods will be incorporated into such future embodiments. In sum, it should be understood that the application is capable of modification and variation.

(222) As used herein, the adverb “substantially” means that a shape, structure, measurement, quantity, time, etc. may deviate from an exact described geometry, distance, measurement, quantity, time, etc., because of imperfections in materials, machining, manufacturing, transmission of data,

computational speed, etc.

(223) All terms used in the claims are intended to be given their ordinary meanings as understood by those knowledgeable in the technologies described herein unless an explicit indication to the contrary is made herein. In particular, use of the singular articles such as “a,” “the,” “said,” etc. should be read to recite one or more of the indicated elements unless a claim recites an explicit limitation to the contrary.

(224) The Abstract is provided to allow the reader to quickly ascertain the nature of the technical disclosure. It is submitted with the understanding that it will not be used to interpret or limit the scope or meaning of the claims. In addition, in the foregoing Detailed Description, it can be seen that various features are grouped together in various embodiments for the purpose of streamlining the disclosure. This method of disclosure is not to be interpreted as reflecting an intention that the claimed embodiments require more features than are expressly recited in each claim. Rather, as the following claims reflect, inventive subject matter lies in less than all features of a single disclosed embodiment. Thus, the following claims are hereby incorporated into the Detailed Description, with each claim standing on its own as a separately claimed subject matter.

Claims

1. A powered surgical handpiece including a handswitch, the powered surgical handpiece comprising: a mounting base defining a pivot axis; an elongated lever defining a lever axis normal to at least a direction of the pivot axis and the lever including: a proximal end pivotably connected to the mounting base at the pivot axis, a distal end opposite the proximal end, a first track on an inner side, and a second track on an outer side opposite the inner side with both tracks parallel to the lever axis, wherein at least one of the first track and the second track is one of a T-slot and a T-track; a spring between the base and the lever biasing the lever about the pivot axis; a run-safe switch slidably disposed on the first track; and a lever extension slidably disposed on the second track.
2. The powered surgical handpiece of claim 1, wherein the first track and the second track are both T-tracks.
3. The powered surgical handpiece of claim 1, wherein the lever has first and second lateral sides connecting the inner and outer sides and the run-safe switch has at least one outboard side extending laterally beyond at least one of the lateral sides.
4. The powered surgical handpiece of claim 3, wherein the outboard side is defined by an indicator arm.
5. The powered surgical handpiece of claim 1, wherein the lever has a slot in the inner side that is in receipt of at least one of a magnet and a magnet retainer fixed to the run-safe switch.
6. The powered surgical handpiece of claim 5, wherein the slot is shorter in length than a length of the lever and the slot defines a travel range of the run-safe switch.
7. The powered surgical handpiece of claim 1, wherein a detent is disposed between the run-safe switch and the lever.
8. The powered surgical handpiece of claim 7, wherein the detent includes a run-safe switch plunger and a first notch associated with an active position of the run-safe switch and a second notch associated with an inactive position of the run-safe switch.
9. The powered surgical handpiece of claim 8, wherein the run-safe switch plunger is disposed in the lever and the notches are disposed on the switch.
10. The powered surgical handpiece of claim 1, wherein a detent is disposed between the lever and the lever extension.
11. The powered surgical handpiece of claim 10, wherein the detent includes an extension plunger and first, second, third and fourth notches respectively associated with first, second, third and fourth positions of the extension.

12. The powered surgical handpiece of claim 11, wherein the extension plunger is disposed in the lever and the notches are disposed in the lever extension.
 13. The powered surgical handpiece of claim 1, wherein the mounting base includes an annular collar that defines a mounting axis normal to at least the direction of the pivot axis and the collar is sized for slideable disposition over a proximal end of the handpiece.
 14. The powered surgical handpiece of claim 13, further comprising: the annular collar defining a positioning tab extending axially in a distal direction and the tab including an angled wedge side; and one of a detent plunger and a detent notch defining a biasing surface disposed on an inner diameter of the annular collar.
 15. The powered surgical handpiece of claim 14, wherein the collar includes at least a second positioning tab, the second positioning tab also extending axially in the distal direction and including an angled wedge side.
 16. The powered surgical handpiece of claim 15, wherein the detent plunger is disposed in the collar.
 17. A handswitch for a powered surgical handpiece, the handswitch comprising: a mounting base defining a pivot axis; an elongated lever defining a lever axis normal to at least a direction of the pivot axis and the lever including: a proximal end pivotably connected to the mounting base at the pivot axis, a distal end opposite the proximal end, a first track on an inner side, and a second track on an outer side opposite the inner side with both tracks parallel to the lever axis, wherein at least one of the first track and the second track is one of a T-slot and a T-track; a spring between the base and the lever biasing the lever about the pivot axis; a run-safe switch slidably disposed on the first track; and a lever extension slidably disposed on the second track.
 18. The handswitch of claim 17, wherein the mounting base includes an annular collar that defines a mounting axis normal to at least the direction of the pivot axis and the collar is sized for slideable disposition over a proximal end of the handpiece.
 19. The handswitch of claim 18, further comprising: the annular collar defining a positioning tab extending axially in a distal direction and the tab including an angled wedge side; and one of a detent plunger and a detent notch defining a biasing surface disposed on an inner diameter of the annular collar.
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